

Physics 7501 Notes

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1 Linear Algebra Review

1.1 Vector Spaces and Hilbert Spaces

Definition 1.1. A *vector space* over a field $\mathbb{F}$ is a set V equipped with two binary operations: addition and scalar multiplication, and for which the following properties hold

- (1) Closure under addition: $|u\rangle + |v\rangle \in V$, $\forall |u\rangle, |v\rangle \in V$
- (2) Closure under scalar multiplication: $\alpha |v\rangle \in V$, $\forall \alpha \in \mathbb{F}, |v\rangle \in V$
- (3) Additive identity: $\exists |0\rangle \in V$ such that $|v\rangle + |0\rangle = |v\rangle$, $\forall |v\rangle \in V$
- (4) Additive inverses: $\exists |u\rangle \in V$ such that $|v\rangle + |u\rangle = |0\rangle$, $\forall |v\rangle \in V$
- (5) Multiplicative scalar identity: $\exists 1 \in \mathbb{F}$ such that $1 |v\rangle = |v\rangle$, $\forall |v\rangle \in V$
- (6) Associativity of addition: $(|u\rangle + |v\rangle) + |w\rangle = |u\rangle + (|v\rangle + |w\rangle)$, $\forall |u\rangle, |v\rangle, |w\rangle \in V$
- (7) Commutativity of addition: $|u\rangle + |v\rangle = |v\rangle + |u\rangle$, $\forall |u\rangle, |v\rangle \in V$
- (8) Distributivity of over vector addition: $\alpha(|u\rangle + |v\rangle) = \alpha |u\rangle + \alpha |v\rangle$, $\forall \alpha \in \mathbb{F}$ and $|u\rangle, |v\rangle \in V$
- (9) Distributivity over scalar addition: $(\alpha + \beta) |v\rangle = \alpha |v\rangle + \beta |v\rangle$, $\forall \alpha, \beta \in \mathbb{F}, |v\rangle \in V$
- (10) Compatibility of scalar multiplication: $\alpha(\beta |v\rangle) = (\alpha\beta) |v\rangle$, $\forall \alpha, \beta \in \mathbb{F}, |v\rangle \in V$

Definition 1.2. A *Hilbert space* is a vector space over $\mathbb{C}$ that is equipped with an inner product $\langle u|v\rangle \in \mathbb{C}$ with the following properties

- (1) Conjugate symmetry: $\langle u|v\rangle = \langle v|u\rangle^*$
- (2) Linearity in the ket: $\langle u|(a|v\rangle + b|w\rangle) = a\langle u|v\rangle + b\langle u|w\rangle$
- (3) Positive definiteness: $\langle v|v\rangle \geq 0$, $\langle v|v\rangle = 0 \iff |v\rangle = |0\rangle$

and is also complete (every Cauchy sequence converges).

Examples of Hilbert spaces:

- (i) n -dimensional Hilbert space on complex numbers: $V \simeq \mathbb{C}^n$
- (ii) $n \times n$ complex matrices equipped with the Hilbert-Schmidt inner product forms a Hilbert space $L_{\mathbb{C}^n}$ (see Homework 1)
- (iii) n -dimensional Hilbert space on real numbers: $V \simeq \mathbb{R}^n$

Examples of Non-Hilbert spaces:

- (i) n -dimensional vector space $V \simeq \mathbb{Q}^n$ on rational numbers is not a Hilbert space because it is not complete, since a Cauchy sequence of rational numbers can converge to an irrational number.

- (ii) The inner product space of continuous square-integrable complex functions on interval $[a, b]$ with the following L^2 inner product:

$$\langle f_1 | f_2 \rangle \equiv \int_a^b f_1^*(x) f_2(x) dx \quad (1.39)$$

is not a Hilbert space due to its incompleteness: a Cauchy sequence of continuous functions can be discontinuous. Its completion is the Hilbert space $L^2[a, b]$.

- (iii) For the same reason of incompleteness, the inner product space of polynomials on $[a, b]$ with the same L^2 inner product is not a Hilbert space, even though it's a subspace of $L^2[a, b]$.

1.2 Cauchy-Schwartz and Triangle Inequality

Theorem 1.1. Cauchy-Schwartz Inequality: $\langle v | v \rangle \langle u | u \rangle \geq |\langle u | v \rangle|^2$

Proof. If $|u\rangle = |0\rangle$, both sides are zero. Otherwise, define

$$|w\rangle = |v\rangle - \lambda |u\rangle, \quad \lambda = \frac{\langle u | v \rangle}{\langle u | u \rangle}.$$

By positivity of the inner product,

$$\begin{aligned} 0 \leq \langle w | w \rangle &= (\langle v | - \lambda^* \langle u |)(|v\rangle - \lambda |u\rangle) \\ &= \langle v | v \rangle - \lambda \langle v | u \rangle - \lambda^* \langle u | v \rangle + |\lambda|^2 \langle u | u \rangle. \end{aligned}$$

Using $\langle v | u \rangle = \langle u | v \rangle^*$ and substituting λ ,

$$\begin{aligned} 0 &\leq \langle v | v \rangle - \frac{|\langle u | v \rangle|^2}{\langle u | u \rangle} - \frac{|\langle u | v \rangle|^2}{\langle u | u \rangle} + \frac{|\langle u | v \rangle|^2}{\langle u | u \rangle} \\ &= \langle v | v \rangle - \frac{|\langle u | v \rangle|^2}{\langle u | u \rangle}. \end{aligned}$$

Multiplying by $\langle u | u \rangle > 0$ gives the inequality. Equality holds precisely when $\langle w | w \rangle = 0$, which, by positive definiteness, means

$$|w\rangle = |0\rangle \iff |v\rangle = \lambda |u\rangle.$$

Theorem 1.2. Triangle Inequality: $\| |v\rangle + |w\rangle \| \leq \| |v\rangle \| + \| |w\rangle \|$

Proof. Expand the square of the norm:

$$\begin{aligned} \| |v\rangle + |w\rangle \|^2 &= (\langle v | + \langle w |)(|v\rangle + |w\rangle) \\ &= \langle v | v \rangle + \langle w | w \rangle + \langle v | w \rangle + \langle w | v \rangle \\ &= \| |v\rangle \|^2 + \| |w\rangle \|^2 + 2 \operatorname{Re} \langle v | w \rangle. \end{aligned}$$

Since $\operatorname{Re}(z) \leq |z|$, and Cauchy-Schwarz gives

$$|\langle v | w \rangle| \leq \sqrt{\langle v | v \rangle \langle w | w \rangle} = \| |v\rangle \| \| |w\rangle \|,$$

we obtain

$$\begin{aligned} \| |v\rangle + |w\rangle \|^2 &\leq \| |v\rangle \|^2 + \| |w\rangle \|^2 + 2 \| |v\rangle \| \| |w\rangle \| \\ &= (\| |v\rangle \| + \| |w\rangle \|)^2. \end{aligned}$$

Taking the non-negative square root proves the triangle inequality.

1.3 Hilbert-Schmidt Inner Product

Definition 1.3. The *Hilbert-Schmidt Inner Product* is a binary operation on $\mathbb{L}_V \times \mathbb{L}_V \rightarrow \mathbb{C}$ defined by

$$(A, B) \equiv \text{tr}(A^\dagger B) \quad A, B \in \mathbb{L}_V$$

Proof. To show that this is indeed an inner product, we need the following properties:

(1) Positive definiteness:

$$(A, A) = \text{tr}(A^\dagger A) = \sum_{i=1}^n \sum_{j=1}^n (A^\dagger)_{ij} A_{ji} = \sum_{i=1}^n \sum_{j=1}^n |A_{ji}|^2 \geq 0,$$

and

$$\sum_{i=1}^n \sum_{j=1}^n |A_{ji}|^2 = 0 \iff A_{ji} = 0 \forall i, j \implies A = 0.$$

(2) Linearity in the second argument:

$$\begin{aligned} (A, \alpha B + \beta C) &= \text{tr}(A^\dagger(\alpha B + \beta C)) \\ &= \text{tr}(\alpha A^\dagger B + \beta A^\dagger C) \\ &= \alpha \text{tr}(A^\dagger B) + \beta \text{tr}(A^\dagger C) \\ &= \alpha(A, B) + \beta(A, C). \end{aligned}$$

(3) Conjugate linearity in first argument / conjugate symmetry:

$$\begin{aligned} (\alpha A + \beta B, C) &= \text{tr}((\alpha A + \beta B)^\dagger C) \\ &= \text{tr}(\alpha^* A^\dagger C + \beta^* B^\dagger C) \\ &= \alpha^*(A, C) + \beta^*(B, C). \end{aligned}$$

$$\begin{aligned} (A, B) &= \text{tr}(A^\dagger B) = \text{tr}((B^\dagger A)^\dagger) \\ &= (\text{tr}(B^\dagger A))^* = (B, A)^*. \end{aligned}$$

Under the Hilbert-Schmidt inner product, the 3 Pauli matrices $\sigma_x, \sigma_y, \sigma_z$

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

and the 2×2 identity matrix $\hat{1}$ together can form a complete orthonormal basis for the vector space $L_{\mathbb{C}^2}$ of all 2×2 complex matrices.

Proof. We know that

$$\sigma_x \sigma_y = i\sigma_z, \quad \sigma_y \sigma_z = i\sigma_x, \quad \sigma_z \sigma_x = i\sigma_y.$$

$\implies$

$$\text{tr}(\sigma_x \sigma_y) = i \text{tr}(\sigma_z) = 0,$$

$$\text{tr}(\sigma_y \sigma_z) = i \text{tr}(\sigma_x) = 0,$$

$$\text{tr}(\sigma_z \sigma_x) = i \text{tr}(\sigma_y) = 0.$$

Thus under the Hilbert-Schmidt inner product, the Pauli matrices are mutually orthogonal. They are also orthogonal to the identity because $(I, \sigma_i) = \text{tr}(\sigma_i) = 0 \implies$ they form a basis for $L_{\mathbb{C}^2}$ since they are a set of 4 linearly independent vectors that span the space (since $\dim(L_{\mathbb{C}^2}) = 4$). To obtain an orthonormal basis, note that $\hat{1}^\dagger \hat{1} = \hat{1}$ and $\sigma_i^\dagger \sigma_i = \hat{1}$. Therefore

$$(I, I) = \text{tr}(I) = 2, \quad (\sigma_i, \sigma_i) = \text{tr}(I) = 2.$$

Each basis matrix has norm $\sqrt{2}$, so the complete orthonormal basis is

$$\left\{ \frac{I}{\sqrt{2}}, \frac{\sigma_x}{\sqrt{2}}, \frac{\sigma_y}{\sqrt{2}}, \frac{\sigma_z}{\sqrt{2}} \right\}.$$

1.4 Properties of the Trace Operation

Some facts about the trace operation

- **Lemma 1.1.** The trace operation is linear over complex scalar $\text{tr}(\alpha A + \beta B) = \alpha \text{tr}(A) + \beta \text{tr}(B)$
- **Lemma 1.2.** The trace operation of a product of linear operators is invariant under cyclic permutations i.e.

Add more lemmas about trace

2 Linear Operators and Observables

Definition 2.1. The *outer product* of two vectors $|u\rangle$ and $|v\rangle$ is the operator $|u\rangle\langle v|$. It acts on any vector $|w\rangle$ by

$$(|u\rangle\langle v|)|w\rangle = \langle v|w\rangle |u\rangle.$$

That is, it computes the overlap with $|v\rangle$, then multiplies $|u\rangle$ by that scalar. Note that the trace of the outer product is equal to the inner product i.e. $\text{tr}(|u\rangle\langle v|) = \langle u|v\rangle$. In the case where $|u\rangle = |v\rangle$ and $|u\rangle$ is normalized i.e. $\langle u|u\rangle = 1$, then $|u\rangle\langle u|$ is the projection operator onto the direction of $|u\rangle$

Definition 2.2. A *linear operator* $\hat{O}$ as a function that maps the vector space $\mathbb{V}$ to itself which is linear in its inputs:

$$\hat{O} \left(\sum_i a_i |v_i\rangle \right) = \sum_i a_i \hat{O}(|v_i\rangle)$$

Assuming $\{|1\rangle, |2\rangle, \dots, |n\rangle\}$ is complete orthonormal basis for V , then each linear operator has a matrix representation

$$\hat{O} = \sum_{i=1}^n \sum_{j=1}^n O_{ij} |i\rangle \langle j| \quad , \quad O_{ij} = \langle i | \hat{O} | j \rangle$$

In other words, a linear operator is defined by its action on a basis. If

$$|v\rangle = \sum_i^n v_i |i\rangle$$

then

$$\begin{aligned} \hat{O} |v\rangle &= \sum_{i=1}^n \sum_{j=1}^n (O_{ij} |i\rangle \langle j|) |v\rangle \\ &= \sum_{i=1}^n \sum_{j=1}^n O_{ij} (v_j |i\rangle) \end{aligned}$$

Not also that

$$\left(\sum_i |i\rangle \langle i| \right) |v\rangle = \sum_i v_i |i\rangle = \hat{1} \quad \implies \quad \sum_i |i\rangle \langle i| = \hat{1}$$

We can define the product of two operators $\hat{M}, \hat{N} \in \mathbb{L}_V$ as

$$\hat{M}\hat{N} = \sum_{i,j,k} M_{ij} N_{jk} |i\rangle \langle k|.$$

Definition 2.3. The *commutator* of two linear operators $\hat{M}, \hat{N} \in \mathbb{L}_V$ is

$$[\hat{M}, \hat{N}] = \hat{M}\hat{N} - \hat{N}\hat{M},$$

and the *anticommutator* is

$$\{\hat{M}, \hat{N}\} = \hat{M}\hat{N} + \hat{N}\hat{M}.$$

Definition 2.4. The *Hermitian conjugate*, or adjoint, of an operator is

$$\hat{O}^\dagger = \sum_{i,j} O_{ji}^* |i\rangle \langle j|.$$

In matrix notation,

$$O^\dagger = (O^T)^* = (O^*)^T, \quad (O^\dagger)_{ij} = O_{ji}^*.$$

This follows from conjugating the coefficient and reversing the outer product:

$$(O_{ij} |i\rangle \langle j|)^\dagger = O_{ij}^* |j\rangle \langle i|.$$

In other words, $\hat{O}^\dagger$ is the operator that satisfies $\langle u | \hat{O} v \rangle = \langle \hat{O}^\dagger u | v \rangle$ for all $|u\rangle$ and $|v\rangle$.

Some facts about Hermitian conjugates, trace, etc.

- **Lemma 2.1.** For two linear operators $\hat{A}$ and $\hat{B}$,

$$(\hat{A}\hat{B})^\dagger = ((\hat{A}\hat{B})^*)^T = (\hat{A}^*\hat{B}^*)^T = (\hat{B}^*)^T(\hat{A}^*)^T = \hat{B}^\dagger\hat{A}^\dagger.$$

Thus, if both $\hat{A}$ and $\hat{B}$ are Hermitian, then $(\hat{A}\hat{B})^\dagger = \hat{B}\hat{A} \implies$ their product is Hermitian if and only if they commute:

- **Lemma 2.2.** The Hermitian adjoint is linear on vectors and conjugate linear over complex scalars i.e. $\hat{A}^\dagger(\alpha|u\rangle + \beta|v\rangle) = \alpha\hat{A}^\dagger|u\rangle + \beta\hat{A}^\dagger|v\rangle$. and $(\alpha\hat{A} + \beta\hat{B})^\dagger = \alpha^*\hat{A}^\dagger + \beta^*\hat{B}^\dagger$.

Add more lemmas

Definition 2.5. An operator $\hat{O}$ is **Hermitian** if and only if $\hat{O} = \hat{O}^\dagger$. Every physical observable in quantum mechanics corresponds to a Hermitian operator.

Every Hermitian operator can be diagonalized with respect to an orthonormal eigenbasis $\{|i\rangle\}$

$$\hat{O} = \sum_i O_i |i\rangle \langle i|$$

This is called the **spectral decomposition** of $\hat{O}$, and the proof follows from the following theorems.

Theorem 2.1. Spectral Decomposition Theorem A linear operator $\hat{O}$ on V is diagonal with respect to some orthonormal basis of V if and only if $\hat{O}$ is normal, i.e. $\hat{O}^\dagger\hat{O} = \hat{O}\hat{O}^\dagger$

Theorem 2.2. A normal matrix is Hermitian if and only if it has real eigenvalues

Theorem 2.3. Eigenkets corresponding to different eigenvalues of a Hermitian matrix are orthogonal

Theorem 2.4. A positive linear operator must be Hermitian. Here positive means

$$\langle v|\hat{O}|v\rangle \geq 0$$

for every $|v\rangle \in V$. In particular, every such expectation value is real. Hermiticity is not assumed in this definition.

Theorem 2.5. Simultaneous diagonalization theorem Two Hermitian linear operators $\hat{A}$ and $\hat{B}$ can be diagonalized simultaneously if and only if

$$[\hat{A}, \hat{B}] = 0.$$

In other words, there is a complete orthonormal basis of simultaneous eigenkets:

$$\hat{A}|a, b\rangle = a|a, b\rangle, \quad \hat{B}|a, b\rangle = b|a, b\rangle. \quad (1.55)$$

3 Projective Measurements

Definition 3.1. A **projection operator** or **projector** is a linear operator $\hat{P} = \hat{P}^2$. For our purposes, we also require that $\hat{P} = \hat{P}^\dagger$ is Hermitian. In other words, the range and kernel of $\hat{P}$ are orthogonal.

Lemma 3.1. A Hermitian operator $\hat{M}$ can be diagonalized with respect to an orthonormal eigenbasis $\{|j\rangle\}$ such that

$$\hat{M} = \sum_j \lambda_j |j\rangle \langle j|.$$

If we use the set $\{m\}$ to label all different (real) eigenvalues of matrix $\hat{M}$, we can rewrite the above diagonal form as

$$\hat{M} = \sum_m m \hat{P}_m, \quad \hat{P}_m = \sum_j \delta_{\lambda_j, m} |j\rangle \langle j|.$$

In other words, $\hat{P}_m$ projects onto the subspace spanned by the eigenvectors that share the same eigenvalue.

Definition 3.2. The *expectation value* $\langle \hat{M} \rangle$ of an operator $\hat{M}$ with respect to a state $|\psi\rangle$ is

$$\langle \hat{M} \rangle \equiv \frac{\langle \psi | \hat{M} | \psi \rangle}{\langle \psi | \psi \rangle}$$

A consequence of this is the **Born Rule**, which states that for a physical observable represented by the Hermitian operator $\hat{M}$ and a state $|\psi\rangle$. Thus the probability of a measurement producing the eigenvalue λ_i is

$$\mathbf{P}(\lambda_i) = \langle \psi | \hat{P}_i | \psi \rangle = \langle \psi | \hat{P}_i^2 | \psi \rangle = \langle \hat{P}_i \psi | \hat{P}_i \psi \rangle = \left\| \hat{P}_i | \psi \rangle \right\|^2$$

Furthermore, the state collapses into a new state

$$\frac{\hat{P}_i | \psi \rangle}{\sqrt{\langle \psi | \hat{P}_i | \psi \rangle}}$$

after measurement.

4 Qubit and Bloch Sphere

Definition 4.1. A *qubit* is a two state system associated with Hilbert space $\mathbb{C}^2$. Most commonly, we can think of it as the space of spin states for an electron. The conventions $|0\rangle$ and $|\uparrow\rangle$ are used to denote the spin up state i.e $s = \frac{1}{2}\hbar$. $|1\rangle$ and $|\downarrow\rangle$ are used to denote the spin down state i.e. $s = -\frac{1}{2}\hbar$ state.

5 Composite Quantum Systems

5.1 Tensors

Definition 5.1. Give a vector space V of dimension m and a vector space W of dimension n , their *tensor product space* is the vector space

$$V \otimes W \equiv \text{span}\{|v\rangle \otimes |w\rangle \mid |v\rangle \in V, |w\rangle \in W\}$$

of dimension mn . An element $|\psi\rangle$ of this vector space is called a *tensor* and takes the form

$$|\psi\rangle = \sum_i a_i (|v_i\rangle \otimes |w_i\rangle)$$

The tensor product is **bilinear** e.g.

$$|\psi\rangle = \sum_i a_i (|v_i\rangle \otimes |w_i\rangle) = \sum_i (a_i |v_i\rangle) \otimes |w_i\rangle = \sum_i |v_i\rangle \otimes (a_i |w_i\rangle).$$

The inner product on the tensor product space is defined

$$(\langle v| \otimes \langle w|)(|v'\rangle \otimes |w'\rangle) = \langle v|v'\rangle \langle w|w'\rangle$$

Since the inner product is conjugate-linear in the first argument and linear in the second, if

$$|\Psi\rangle = \sum_i a_i |v_i\rangle \otimes |w_i\rangle \quad , \quad |\Phi\rangle = \sum_j b_j |v'_j\rangle \otimes |w'_j\rangle$$

then

$$\langle \Psi | \Phi \rangle = \sum_{i,j} a_i^* b_j \langle v_i | v'_j \rangle \langle w_i | w'_j \rangle.$$

For linear operators $\hat{A}$ on V and $\hat{B}$ on W , we can construct a linear operator in the tensor product space

$$(\hat{A} \otimes \hat{B})(|v\rangle \otimes |w\rangle) = \hat{A}|v\rangle \otimes \hat{B}|w\rangle$$

5.2 Generalized Measurements

One way we can glean information about a system is to let it interact with another system that we can measure (called the ancilla). Consider two Hilbert spaces $\mathbb{V}_s$ (measured system) and $\mathbb{V}_a$ (ancilla system) and a state $|\Psi\rangle$ in their tensor product space

$$|\Psi\rangle \equiv |\psi\rangle_s \otimes |\phi\rangle_a$$

evolving unitarily under a Hamiltonian $\hat{H}$ for time t . Suppose we perform a projective measurement on $\mathbb{V}_a$ associated with the linear operator $\hat{1} \otimes \hat{P}_a$, where $\hat{P}_a$ has the spectral decomposition

$$\hat{P}_a = \sum_q \lambda_q |q\rangle \langle q|$$

and $\{|q\rangle\}$ forms a complete orthonormal basis for $\mathbb{V}_a$. Immediately before the measurement, the joint state is

$$|\Psi'\rangle = \hat{U}(|\psi\rangle_s \otimes |\phi\rangle_a)$$

where $\hat{U} = e^{-i\hat{H}t/\hbar}$. After such a measurement with measurement outcome λ_q , the state collapses into a new (unnormalized) state $\hat{M}_q(|\psi\rangle_s \otimes |\phi\rangle_a)$ where

$$\hat{M}_q \equiv {}_a\langle q | \hat{U} | \phi \rangle_a \in \mathbb{L}_{\mathbb{V}_s}$$

is a linear operator on $\mathbb{V}_s$. In other words, $\hat{M}_q$ describes the effect on the first system associated with obtaining a measurement outcome λ_q from the ancilla system. More explicitly,

$$\hat{M}_q |\psi\rangle_s = (\hat{1}_s \otimes {}_a\langle q |) \hat{U} (|\psi\rangle_s \otimes |\phi\rangle_a)$$

Reading from right to left, this means that we can obtain $\hat{M}_q |\psi\rangle_s$ by

1. Applying the joint evolution operator $\hat{U}$ to the initial joint state

$$\hat{U}(|\psi\rangle_s \otimes |\phi\rangle_a) = \sum_{i,j} c_{ij}(|i\rangle_s \otimes |j\rangle_a)$$

2. Taking the inner product

$$\left\langle (\hat{1}_s \otimes {}_a\langle q|) \left| \sum_{i,j} c_{ij}(|i\rangle_s \otimes |j\rangle_a) \right. \right\rangle = \sum_{i,j} c_{ij} |i\rangle_s {}_a\langle q|j\rangle_a = \sum_{i,j} c_{ij} \delta_{qj} |i\rangle_s = \sum_i c_{iq} |i\rangle_s$$

Since

$$\begin{aligned} \hat{P}_q |\Psi'\rangle &= \sum_j (\hat{M}_j |\psi\rangle_s) \otimes (|q\rangle_a {}_a\langle q|j\rangle_a) \\ &= \sum_j \delta_{qj} (\hat{M}_j |\psi\rangle_s) \otimes |q\rangle_a \\ &= \hat{M}_q (|\psi\rangle_s \otimes |q\rangle_a) \end{aligned}$$

by the Born rule, the probability that a measurement on $\mathbb{V}_a$ produces q is

$$\mathbf{P}(q) = \langle \Psi' | \hat{P}_q | \Psi' \rangle = {}_s\langle \psi | \hat{M}_q^\dagger \hat{M}_q | \psi \rangle_s$$

thus the $\hat{M}_q$'s satisfy the completeness relation $\sum_q \hat{M}_q^\dagger \hat{M}_q = \hat{1}_s$. Alternatively, we can also write

$$\mathbf{P}(q) = \text{tr}(\hat{M}_q |\psi\rangle_s \langle \psi|_s \hat{M}_q^\dagger)$$

Example 5.1. Add examples for clarity

6 Appendix, Proofs of Theorems and Lemmas

Theorem 6.1. Spectral Decomposition Theorem A linear operator $\hat{O}$ on V is diagonal with respect to some orthonormal basis of V if and only if $\hat{O}$ is normal, i.e. $\hat{O}^\dagger \hat{O} = \hat{O} \hat{O}^\dagger$

Proof. Being diagonal in an orthonormal basis means that

$$\hat{O} = \sum_i O_i |i\rangle \langle i|, \quad (1.53)$$

where the $|i\rangle$ form a complete orthonormal basis and

$$\hat{O} |i\rangle = O_i |i\rangle.$$

Proof: an orthonormal eigenbasis implies normality. Suppose $\hat{O}$ has the form (1.53). Taking its adjoint gives

$$\hat{O}^\dagger = \sum_i O_i^* |i\rangle \langle i|.$$

Using $\langle i|j\rangle = \delta_{ij}$,

$$\begin{aligned} \hat{O}^\dagger \hat{O} &= \sum_{i,j} O_i^* O_j |i\rangle \langle i|j\rangle \langle j| \\ &= \sum_i |O_i|^2 |i\rangle \langle i|. \end{aligned}$$

Similarly,

$$\hat{O}\hat{O}^\dagger = \sum_i |O_i|^2 |i\rangle\langle i|.$$

Therefore $\hat{O}^\dagger\hat{O} = \hat{O}\hat{O}^\dagger$. Proof: normality implies an orthonormal eigenbasis. We first establish a useful fact. If $\hat{O}$ is normal and

$$\hat{O}|v\rangle = \lambda|v\rangle,$$

then

$$\hat{O}^\dagger|v\rangle = \lambda^*|v\rangle.$$

To prove this, define

$$\hat{A} = \hat{O} - \lambda\hat{1}.$$

Normality of $\hat{O}$ implies normality of $\hat{A}$, because

$$\hat{A}^\dagger\hat{A} - \hat{A}\hat{A}^\dagger = \hat{O}^\dagger\hat{O} - \hat{O}\hat{O}^\dagger = 0.$$

Consequently,

$$\begin{aligned}\|\hat{A}^\dagger|v\rangle\|^2 &= \langle v|\hat{A}\hat{A}^\dagger|v\rangle \\ &= \langle v|\hat{A}^\dagger\hat{A}|v\rangle \\ &= \|\hat{A}|v\rangle\|^2 = 0.\end{aligned}$$

Thus $\hat{A}^\dagger|v\rangle = 0$, proving the fact. We now use induction on $n = \dim V$. The result is immediate for $n = 1$. For $n > 1$, a complex matrix has at least one eigenvalue, so choose a normalized eigenvector $|1\rangle$:

$$\hat{O}|1\rangle = \lambda_1|1\rangle, \quad \hat{O}^\dagger|1\rangle = \lambda_1^*|1\rangle.$$

Consider its orthogonal complement

$$W = \{|w\rangle \in V : \langle 1|w\rangle = 0\}.$$

For any $|w\rangle \in W$,

$$\langle 1|\hat{O}|w\rangle = \lambda_1\langle 1|w\rangle = 0,$$

where we used the adjoint of $\hat{O}^\dagger|1\rangle = \lambda_1^*|1\rangle$. Likewise,

$$\langle 1|\hat{O}^\dagger|w\rangle = \lambda_1^*\langle 1|w\rangle = 0.$$

Therefore both $\hat{O}$ and $\hat{O}^\dagger$ map W into itself. Their restrictions to W still commute, so the restriction of $\hat{O}$ is normal. Since $\dim W = n - 1$, the induction hypothesis supplies an orthonormal eigenbasis

$$|2\rangle, \dots, |n\rangle$$

for W . Together with $|1\rangle$, these form a complete orthonormal eigenbasis for V . Hence

$$\hat{O} = \sum_i \lambda_i |i\rangle\langle i|$$

Theorem 6.2. A normal matrix is Hermitian if and only if it has real eigenvalues

Proof. By Theorem 2.1, a normal operator has a spectral decomposition

$$\hat{O} = \sum_i \lambda_i |i\rangle\langle i|, \quad \hat{O}^\dagger = \sum_i \lambda_i^* |i\rangle\langle i|.$$

If $\hat{O}$ is Hermitian, then

$$\lambda_i = \langle i|\hat{O}|i\rangle = \langle i|\hat{O}^\dagger|i\rangle = \lambda_i^*,$$

so every eigenvalue is real. Conversely, if all λ_i are real, then

$$\hat{O}^\dagger = \sum_i \lambda_i^* |i\rangle\langle i| = \sum_i \lambda_i |i\rangle\langle i| = \hat{O}.$$

Thus $\hat{O}$ is Hermitian.

Theorem 6.3. Eigenkets corresponding to different eigenvalues of a Hermitian matrix are orthogonal

Proof. Let $\hat{M} = \hat{M}^\dagger$, and suppose

$$\hat{M}|i\rangle = \lambda_i|i\rangle, \quad \hat{M}|j\rangle = \lambda_j|j\rangle, \quad \lambda_i \neq \lambda_j.$$

A Hermitian operator is normal, so Theorem 2.2 tells us that its eigenvalues are real. Taking the adjoint of the first eigenvalue equation gives

$$\langle i|\hat{M} = \lambda_i\langle i|.$$

Now evaluate the same matrix element in two ways:

$$\langle i|\hat{M}|j\rangle = \lambda_j\langle i|j\rangle$$

by acting to the right, while

$$\langle i|\hat{M}|j\rangle = \lambda_i\langle i|j\rangle$$

by acting to the left. Subtracting,

$$(\lambda_i - \lambda_j)\langle i|j\rangle = 0.$$

Since $\lambda_i \neq \lambda_j$,

$$\langle i|j\rangle = 0$$

For equal eigenvalues, this argument imposes no condition on the overlap. Vectors within a degenerate eigenspace need not initially be orthogonal, but an orthonormal basis can be chosen there.

Theorem 6.4. A positive linear operator must be Hermitian. Here positive means

$$\langle v|\hat{O}|v\rangle \geq 0$$

for every $|v\rangle \in V$. In particular, every such expectation value is real. Hermiticity is not assumed in this definition.

Proof. Decompose $\hat{O}$ into its Hermitian and anti-Hermitian parts:

$$\hat{O} = \hat{H} + i\hat{K},$$

where

$$\hat{H} = \frac{\hat{O} + \hat{O}^\dagger}{2}, \quad \hat{K} = \frac{\hat{O} - \hat{O}^\dagger}{2i}.$$

Directly taking adjoints shows

$$\hat{H}^\dagger = \hat{H}, \quad \hat{K}^\dagger = \hat{K}.$$

For every $|v\rangle$,

$$\langle v|\hat{O}|v\rangle = \langle v|\hat{H}|v\rangle + i\langle v|\hat{K}|v\rangle.$$

Both expectations on the right are real because $\hat{H}, \hat{K}$ are Hermitian. Positivity requires the left side to be real, so its imaginary part vanishes:

$$\langle v|\hat{K}|v\rangle = 0 \quad \text{for every } |v\rangle.$$

Since $\hat{K}$ is Hermitian, it has a complete orthonormal eigenbasis. For any normalized eigenvector $|i\rangle$,

$$\hat{K}|i\rangle = k_i|i\rangle \implies k_i = \langle i|\hat{K}|i\rangle = 0.$$

All eigenvalues vanish, hence $\hat{K} = 0$. Therefore

$$\boxed{\hat{O} = \hat{H} = \hat{O}^\dagger.}$$

In fact, this proof shows that real expectation values for every vector already imply Hermiticity. Nonnegativity further ensures that the eigenvalues are nonnegative.

Theorem 6.5. Simultaneous diagonalization theorem Two Hermitian linear operators $\hat{A}$ and $\hat{B}$ can be diagonalized simultaneously if and only if

$$\boxed{[\hat{A}, \hat{B}] = 0.}$$

In other words, there is a complete orthonormal basis of simultaneous eigenkets:

$$\hat{A}|a, b\rangle = a|a, b\rangle, \quad \hat{B}|a, b\rangle = b|a, b\rangle. \quad (1.55)$$

If several independent states have the same pair of eigenvalues, an additional label is needed. Proof: simultaneous diagonalization implies commutation. Using conventional basis indices, suppose

$$\hat{A}|i\rangle = a_i|i\rangle, \quad \hat{B}|i\rangle = b_i|i\rangle$$

for every vector in a complete basis. Then

$$\hat{A}\hat{B}|i\rangle = a_i b_i|i\rangle = \hat{B}\hat{A}|i\rangle.$$

Thus

$$[\hat{A}, \hat{B}]\lvert i\rangle = 0$$

for every basis vector. By linearity, the commutator vanishes on every vector:

$$[\hat{A}, \hat{B}] = 0.$$

Proof. Commutation implies simultaneous diagonalization. Suppose $[\hat{A}, \hat{B}] = 0$. Since $\hat{A}$ is Hermitian, V decomposes into its mutually orthogonal eigenspaces. Let V_a be the eigenspace with eigenvalue a . For any $|v\rangle \in V_a$,

$$\hat{A}|v\rangle = a|v\rangle.$$

Using commutation,

$$\begin{aligned}\hat{A}(\hat{B}|v\rangle) &= \hat{B}\hat{A}|v\rangle \\ &= \hat{B}(a|v\rangle) \\ &= a(\hat{B}|v\rangle).\end{aligned}$$

Therefore $\hat{B}|v\rangle$ also belongs to V_a : $\hat{B}$ preserves every eigenspace of $\hat{A}$. Restrict $\hat{B}$ to one such eigenspace V_a . This restriction is still Hermitian, so it has an orthonormal eigenbasis within V_a . Every vector in that basis is: - an eigenvector of $\hat{A}$, because it belongs to V_a ; - an eigenvector of $\hat{B}$, by construction. Perform this diagonalization separately in every eigenspace of $\hat{A}$. Combining the resulting bases gives a complete orthonormal basis of simultaneous eigenvectors for V . $\square$ The degenerate case is the important step: commutation does not guarantee that every arbitrarily chosen eigenbasis of $\hat{A}$ already diagonalizes $\hat{B}$. It guarantees that we can choose suitable vectors within each degenerate eigenspace. The spectral-projector form at the bottom of the page A Hermitian operator has an orthonormal eigenbasis in which

$$\hat{M} = \sum_i \lambda_i |i\rangle\langle i|. \quad (1.56)$$

If m labels the distinct eigenvalues, group all basis vectors sharing each eigenvalue:

$$\boxed{\hat{M} = \sum_m m \hat{P}_m, \quad \hat{P}_m = \sum_j \delta_{\lambda_j, m} |j\rangle\langle j|}.$$

The operator $\hat{P}_m$ projects onto the entire eigenspace with eigenvalue m . Indeed,

$$\begin{aligned}\sum_m m \hat{P}_m &= \sum_m \sum_j m \delta_{\lambda_j, m} |j\rangle\langle j| \\ &= \sum_j \lambda_j |j\rangle\langle j| \\ &= \hat{M}.\end{aligned}$$

For each fixed j , only the term $m = \lambda_j$ survives. This form makes degeneracy explicit without requiring a particular basis inside each eigenspace.